STANDATA variance 23-BCV-008

Building

Residential exhaust outlet clearance from air intakes

Date Issued: June 2024

Purpose

To provide an alternative solution to clearance requirements between domestic kitchen, bathroom and clothes dryer exhaust outlets from air intakes.

Discussion

The requirements for exhaust clearances from soffits were updated in the National Building Code of Canada 2020 and subsequently harmonized within the National Building Code - 2023 Alberta Edition (NBC(AE)). The requirements prescribed a clearance of 1800 mm from exhaust vents below soffits or to install solid soffits where the clearance is less than 1800 mm below soffits.

Clearance was increased from 900 mm to 1800 mm for clothes dryer exhaust vents to air intakes. This change did not differentiate between clothes dryers in common laundry rooms serving residential buildings or domestic clothes dryers serving a single dwelling unit or a house with a secondary suite.

Clearance of 1800 mm was also added for bathroom ventilation exhaust from air intakes.

The rationale presented within the requirements for exhaust clearances explained the application to soffits but did not provide a rationale applicable to clothes dryer exhaust or bathroom exhaust from air intakes. The rationale also describes “air containing moisture rises from the exhaust outlets” as part of the problem. The rationale was silent on kitchen exhaust from dwelling units. But to clarify are also addressing kitchen exhaust outlets.

Kitchen, bathroom, and clothes dryer exhaust serving individual dwelling units or houses with a secondary suite typically operate intermittently.

CSA-B149.1 “Natural gas and propane installation code” (CSA B149.1) provides for vent terminations from building air intakes prescribing a clearance of 900 mm for domestic clothes dryers and 1800 mm for commercial clothes dryers. This supports the allowance for the clearance reduction when dealing with domestic clothes dryers. The domestic clothes dryer exhaust air would typically contain more moisture than bathroom exhaust air.

Code References

Sentences, 9.32.3.13.(4)(5)(6)

9.32.3.13. Outdoor Intake and Exhaust Openings

4) Except as provided in Sentences (5) and (6), exhaust outlets that discharge air containing moisture, such as bathroom ventilation and clothes dryer exhaust outlets, shall be located at least 1 800 mm from air intakes and vented soffits.

©2024 Government of Alberta | June 2024 | Municipal Affairs

Archived

5) Where an exhaust outlet referred to in Sentence (4) is located within a soffit, the soffit shall either be unvented, or if vented, the full depth of the soffit shall be blocked for a distance of 1 800 mm on each side of the exhaust outlet.

6) Where an exhaust outlet referred to in Sentence (4) is located in a sidewall less than 1 800 mm from a soffit, a section of the soffit above the exhaust outlet shall be unvented, or if vented, the full depth of the soffit shall be blocked in accordance with the widths stipulated in Table 9.32.3.13.-A, centred over the location of the outlet.

CSA-B149.1 Sentences 7.5.1 and 7.5.2

7.5 Domestic-type clothes dryers

7.5.1
A dryer shall be equipped with a moisture-exhaust duct that terminates outside the ***building***, and the duct shall be constructed of a material that is
(a) ***noncombustible***; or
(b) ***certified*** as meeting the requirements for Class 1 air ducts contained in CAN/ULC-S110.

7.5.2
A moisture-exhaust duct shall not terminate within 3 ft (1 m) in any direction of any ***pressure regulator vent*** termination or fresh-air intake.

CSA-B149.1 Clause 8.14.8 (c)

8.14.8
A vent shall not terminate
c) within 6 ft (1.8 m) of a mechanical air-supply inlet to any building;

Rationale for Clearance change

Problem

Exhaust from clothes dryers and bathroom fans is allowed to discharge from vents installed in the soffits and in the side walls below the soffits. This warm, moist air can enter the attic space through any vented soffits in the immediate vicinity of the exhaust vent, potentially causing mold, mildew, rot, loss of the insulation's R-value, ice damming, and water infiltration. Approximately 5% of houses with soffit exhaust vents conforming to the current code requirements show signs of mold, mildew, rot, and water infiltration.

Rationale

Installing unvented soffits or blocking a section of vented soffits for a certain width on each side of exhaust vents will prevent the infiltration of warm, moist air from the exhaust vents into the attic space, which can cause mold, mildew, rot, loss of the insulation's R-value, ice damming, and water infiltration.

Air containing moisture rises from the exhaust outlets and infiltrates the attic through the soffits, then condensation builds up in the attic over time, resulting in the growth of mold, which causes the deterioration of the ceiling below the roof. Most homeowners do not realize that the roof sheathing on their house is rotting until they replace the roof or notice mold on an interior finish—mold that could be affecting the health of the occupants.

Archived

©2024 Government of Alberta | June 2024 | Municipal Affairs

This proposed change is based on the provisions of CSA B139, “Installation code for Oil-Burning equipment” which stipulates that side wall vents need to be a minimum 1.8m from soffits or unvented soffits must be installed.

Application

This variance applies to kitchen, bathroom, and clothes dryer exhaust outlets serving individual dwelling units or a house with a secondary suite.

Variance

This variance provides approximately equivalent or greater safety performance with respect to persons and property as provided for by the *Safety Codes Act* and the NBC(AE).

1. The horizontal clearance of kitchen, bathroom and clothes dryer exhaust outlets is permitted to be within 900 mm from air intakes serving the individual dwelling unit or house with a secondary suite.

Note: Air intake is understood to be connected to a fan.

This VARIANCE is applicable throughout the Province of Alberta.

Archived

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code - 2023 Alberta Edition

Issued by the Provincial Building Administrator

[Original signed]

Paul Chang
Provincial Building Administrator

Alberta Municipal Affairs – Technical and Corporate Services
Email: safety.services@gov.ab.ca Phone: 1-866-421-6929

To sign up for our List Subscription Service go to: municipalaffairs.gov.ab.ca/am_list_subscription_services

©2024 Government of Alberta | June 2024 | Municipal Affairs